Remediation Workbench: Technical Report

Center for eLearning · ADA Title II Compliance

INTERNAL DISTRIBUTION ONLY · LOCAL PROCESSING · NO EXTERNAL SERVICES

This report documents the current implementation of the Remediation Workbench web application. It is written to match the original template formatting: header, page numbers, minimal beige table headers, and print-ready margins. All content reflects the updated codebase with no version history, changelog, or knowledge-transfer sections included.

REPORT STRUCTURE

1 Executive summary
2 Architecture
3 Workflow
4 Word
5 PowerPoint
6 PDF routes
7 Colour contrast
8 Alt text & handwriting
9 Logic checks
10 What it cannot do
11 Installation

1. Executive summary

Remediation Workbench is a local web application that makes Microsoft Word (.docx), PowerPoint (.pptx), and PDF files accessible in bulk, on a single computer, without sending files to an outside service. It was built for the Center for eLearning to support ADA Title II and WCAG 2.1 AA remediation for course materials.

The application runs entirely on the operator's machine. A faculty or staff member opens `http://localhost:8080` in a browser, drops in files, and receives remediated copies back with an HTML and CSV report of what was fixed and what still needs a human decision.

KEY DESIGN RULES

- ☐ **Files never leave the computer.** Processing, OCR, and AI alt-text drafting happen locally. No file content is uploaded to a vendor. The only optional network call is the link checker, which fetches public URLs if enabled.
- ☐ **Layout stays the same.** Word and PowerPoint fixes preserve visual layout. PDFs are tagged in place — text is not redrawn or reflowed except for the optional reflowable text version. Verified pixel-by-pixel for born-digital PDFs.
- ☐ **Automatic by default.** In Automatic mode the system applies safe fixes without prompting: titles, language, heading structure, table headers, colour contrast, reading order cleanup, and decorative marking. Manual decisions are limited to alt text, handwriting, and flagged ambiguous cases.

2. Architecture and technology stack

One Docker container serves the web UI on port 8080. A second local service, Ollama, runs a vision model on the host for alt-text and handwriting drafts. Nothing calls outside except the optional link checker. All remediators run inside the container with no cloud dependencies.

LAYER	STACK	PURPOSE
Web server	Python, Flask, Waitress	Pages, uploads, APIs, background job queue (2 workers), Stop / delete / history, alt-text and handwriting APIs
Container	Docker Desktop + WSL2 (Windows)	Packages every tool so the same image runs on Windows, macOS, Linux. Volumes mount `data/jobs` and `data/history`
PDF work	pikepdf, PyMuPDF, pdfminer.six, qpdf	Read and rewrite tags, colours, structure tree, marked content, artifacts, form XObjects, and transparency. No rasterization for tagged route.
Office work	python-docx, python-pptx, LibreOffice	Fix Word / PowerPoint semantics directly; export tagged PDF via LibreOffice headless for Rebuild-through-Word option
OCR	Tesseract 5, OCRmyPDF, OpenCV	Read scans, detect rotation (0/90/180/270), deskew, confidence scoring, create hidden text layer for scanned PDFs
Equations	pandoc, matplotlib (LaTeX)	Convert LaTeX fragments to Word OMML equations when generating typed versions for handwritten pages
AI	Ollama + Qwen2.5-VL 7B on host	Alt-text drafts, handwriting transcription, drawing vs. text classification. GPU accelerated, images never leave host.

FILE MAP — CURRENT LAYOUT

FILE	RESPONSIBILITY
`app.py`	Web routes, job queue, Stop, delete, history retention, alt-text review API, handwriting review API, download, report rendering
`remediate/pipeline.py`	Routes each file by type and detected state (Word, PowerPoint, Tagged PDF, Untagged born-digital, Scanned with hidden OCR layer)
`remediate/common.py`	Options object, remediation report model, export helpers, colour measurement, link checking toggle, language defaults
`remediate/docx_fix.py`	Word fixes: title / language, fake headings → H1–H3, skipped levels repaired, typed bullets optional, first table row as header, legacy .doc → .docx conversion
`remediate/pptx_fix.py`	PowerPoint: title / language, low-contrast colours measured on rendered slides, legacy .ppt → .pptx, theme / picture bullets → standard, background

FILE	RESPONSIBILITY
	pictures → decorative
`remediate/pdf_fix.py`	PDF routing dispatcher: tagged keeper, born-digital tagger, scanned layer builder. Manages artifact conversion and structure tree rebuilds.
`remediate/pdf_tagged.py`	Already-tagged path: preserve tags, inject title / language / display doc title, discover figures even with custom tag names, pre-mark thin lines as decorative
`remediate/pdf_born.py`	Untagged born-digital: tag in place, nothing redrawn, headings / paragraphs / lists / reading order, slide decks H1/H2, LaTeX spaces restored, optional Rebuild through Word
`remediate/pdf_scanned.py`	Scanned with OCR layer: _accessible.pdf looks identical, images untouched as background, invisible tagged text layer, headings from TOC / outline / large type
`remediate/contrast.py`	Colour contrast engine: measures rendered RGB, WCAG 4.5:1 / 3:1, hue-preserving darken / lighten, CMYK / ICC / spot handling
`remediate/alt.py`	Alt-text orchestration, Ollama client, nearby text context, flags for text-in-image, colour-only, long-description candidates
`remediate/handwriting.py`	Handwriting detection, drawing detector for hand-drawn figures, transcription, typed version builder with real equations + tagged PDF

3. End-to-end workflow

Upload → Remediate → Review & download. Designed for one operator to clear a batch of course files without switching tools.

01 · UPLOAD
Drag up to 1000 MB (configurable via MAX_UPLOAD_MB). Accepts .docx, .doc, .pptx, .ppt, .pdf. Duplicate name detection. Files written to data/jobs/<id>/input.

02 · CHOOSE MODE
Automatic (recommended) or Choose options myself. Automatic applies title/language, headings, table headers, contrast, artifacts, reading-order cleanup. Manual exposes toggles for list conversion, tagged PDF rebuild, contrast threshold.

03 · QUEUE
2 files at a time (WORKERS=2). Additional files wait in queue. Stop button cancels current job and preserves partial report. Progress streamed via polling API.

04 · REMEDIATE PIPELINE
Type detection → route selection → fix pass → measurement pass → report generation. Word and PowerPoint fixed in place. PDFs routed to tagged / born-digital / scanned paths.

05 · REVIEW ALT TEXT
Every image appears on alt-text review page with AI draft, nearby caption/paragraph context, and flags. Approve / edit / mark decorative / Approve all / Draft with AI.

06 · HANDWRITING OPTIONAL
If handwriting detected, handwritten pages show side-by-side with typed version containing real equations. Operator approves typed version or keeps original.

07 · DOWNLOAD AND HISTORY: Downloads: remediated file, _accessible.pdf for scanned, _text-version tagged PDF, remediation-report.html/.csv. HISTORY view and RETENTION_DAYS auto-cleanup.

SETTING	DEFAULT	EFFECT
`RETENTION_DAYS`	14	Jobs in `data/jobs/` deleted after N days. History entry remains with counts, not file content.
`HISTORY`	on	Persistent index of completed jobs, searchable by file name, date, operator decisions.
`WORKERS`	2	Parallel files. Higher values increase RAM/CPU, not recommended on laptops with large scans.

4. Remediation logic — Microsoft Word

Layout preserved. All fixes operate on the document model, not by re-exporting to a new layout.

FIXED AUTOMATICALLY

- [ ] Document title and language set from file name and `DOC_LANG` if missing; display language applied to all styles.
- [ ] Fake headings detected by bold / font size / spacing heuristics → converted to real H1–H3 styles with outline levels.
- [ ] Skipped heading levels repaired (e.g., H1 → H3 becomes H1 → H2) without changing visual size where possible.
- [ ] Hand-typed bullets (•, -, o, numbers typed manually) → typed list structure when `CONVERT_LISTS` is enabled.
- [ ] First row of every table marked as header row if not already, when table contains more than one row and first row appears as headers.
- [ ] Legacy `.doc` → `.docx` converted via LibreOffice or python-docx compatibility path to enable accessibility features.
- [ ] Empty paragraphs used for spacing reduced; section spacing preserved via paragraph spacing, not blank lines.

FLAGGED FOR HUMAN REVIEW

- [ ] Images without alt text, images where alt equals file name (e.g., `image1.png`), auto-generated alt from Word, vague alt like “image”.
- [ ] Links with vague text: “click here”, “read more”, raw URLs as display text. Broken links listed when `CHECK_LINKS` enabled.
- [ ] Merged cells, layout tables (tables used for positioning detected by high column count and lack of header-like first row), nested tables.
- [ ] Hand-typed lists that remain ambiguous, empty paragraphs that may convey meaning, floating objects without anchor text.

Report fields: Each flagged item includes page reference, surrounding text snippet, suggested fix, and whether it blocks WCAG 2.1 AA.

5. Remediation logic — PowerPoint

FIXED AUTOMATICALLY

- [ ] Presentation title, language, and reading order for placeholders set; slide master language propagated.
- [ ] Low-contrast colours measured on rendered slides (via python-pptx + rendered colour extraction), not just theme values. Text darkened / lightened keeping hue when Fix low-contrast is ticked (default).
- [ ] Legacy `.ppt` → `.pptx` converted to enable tagging and alt-text storage.
- [ ] Theme / picture bullets → standard paragraph bullets to expose list structure to screen readers.
- [ ] Background pictures and slide master decorative shapes marked as decorative (alt = “”).

FLAGGED FOR HUMAN REVIEW

- [ ] Slides with no title, empty title placeholder, duplicate titles across deck (flagged as “duplicate — consider unique titles”).
- [ ] Pictures without alt, charts without alt, tables without header row, vague or broken links.
- [ ] Busy slides: more than 6 distinct text boxes or more than 40 words in notes may impede comprehension — flagged for simplification.
- [ ] Text on photos that fails contrast, flagged unless see-through backing option is applied.

Optional: see-through backing for text on photos

When enabled, a 84% opaque white backing is added behind text boxes that sit over photographic backgrounds. Preserves image, achieves 4.5:1 without recoloring entire theme. Off by default for Word, optional for PowerPoint.

Optional: tagged PDF export

PowerPoint can export a tagged PDF via LibreOffice. Includes titles, alt text, table headers, and reading order from slide placeholders. Used when distribution requires PDF only.

6. PDF — three routes

PDF handling is split into three distinct routes, selected automatically by inspecting the file's structure, not its file name.

ROUTE	DETECTION	WHAT HAPPENS
Already tagged	Has StructTreeRoot, RoleMap, at least one page with marked content	Tags kept intact; incremental repair only
Untagged born-digital	Extractable text, no StructTreeRoot, not scanned	Tagged in place, nothing redrawn, verified pixel by pixel
Scanned with hidden OCR layer	Page is image-only or image + existing OCR, low extractable text	Creates _accessible.pdf that looks identical + _text-version reflowable

6.1 Already tagged — tags kept

- Title, language, and Display DocTitle flag set from metadata and DOC_LANG if missing. Viewer shows title, not file name.
- Figures discovered even with custom tag names (e.g., `Sect/Figure/CustomRole` mapped via RoleMap). Thin lines (< 2pt) pre-marked as decorative before tagging to avoid “figure” noise.
- Decorative paths, watermarks, and background shadings converted to Artifact (Type / Pagination / Background / Layout). H1 added if document has no H1 but has H2/H3 hierarchy.
- Non-table structures that were incorrectly tagged as Table (e.g., multi-column layout using table tags) lose table tags and become P or Div, preserving reading order.
- Watermarks detected by low opacity, large size, diagonal rotation, or repeated XObject across pages → become Artifact / Watermark.

6.2 Untagged born-digital — tagged in place

- Nothing redrawn. Existing text operators reused, no re-creation of page content streams that would shift glyphs. Verified by rendering before and after and comparing pixel checksums in sampling mode.
- Headings, paragraphs, lists, and reading order inferred from font size, weight, spacing, indentation, and line clustering. Figures tagged from image XObjects. Shapes and running heads/footers → Artifact.
- Slide decks detected by low text per page, large titles, and repeated master background → mapped to H1 (slide title) + H2 (subsections) + P/List.
- LaTeX PDFs: inter-word spaces restored where TeX emitted TJ arrays without explicit spaces. Ligatures (fi, fl) expanded for accessibility but visual remains via ActualText.
- **Rebuild through Word option:** Converts PDF → DOCX via LibreOffice + layout analysis, fixes semantics in Word layer, then exports tagged PDF. Used when source PDF has severe reading-order issues.

6.3 Scanned with hidden OCR layer

- Main output `_accessible.pdf` looks exactly like original. Images untouched as background. Invisible tagged text layer added on top via OCRmyPDF + custom structure insertion, so visual remains identical while screen readers read tagged layer.
- Headings derived from TOC / document outline (if present), or large type size and centered positioning. Running heads/footers/page numbers marked as Artifact / Pagination / Header.
- Sideways pages detected via Tesseract OSD + OpenCV; rotated to upright for text layer but background image rotated to match, so appearance unchanged.
- OCR confidence checked per word. Low-confidence words (confidence < 60) listed in report with page, bounding box, and raw OCR text for human verification.
- Reflowable `_text-version.pdf` also generated: fully tagged, selectable text, linear reading order, suitable for reflow on mobile or for text-to-speech.

7. Colour contrast

With **Fix low-contrast** ticked (default), all text that fails WCAG 2.1 SC 1.4.3 is automatically corrected while preserving hue.

RULE	THRESHOLD	HOW MEASURED / FIXED
Normal text	4.5:1	Measured on rendered RGB after blending transparency. Word and rebuilt PDFs darkened/lightened keeping hue (HSL adjustment). Standard colours used: black, dark blue #0B2A6B, dark red #6E0D0D, dark green #0D4A1A, white.
Large text	3:1	Large defined as 18pt or 14pt bold. Same fixing logic with higher tolerance.
Already-tagged PDFs	Same	Colours rewritten inside PDF content streams. Handles CMYK, ICC profiles, spot colours, rotated/vertical text, form XObjects, and shadings. Gradients replaced with solid meeting ratio.
See-through backing	4.5:1	Optional 84% white backing behind text on photos to avoid recolouring image. Used for PowerPoint text over photos.

- [ ] Handles CMYK, ICC/spot colour, rotated/vertical text, form XObjects, transparency groups, gradients (replaced with solid meeting contrast).
- [ ] Text on photos flagged if backing not enabled; text inside raster images cannot be measured — flagged for alt-text review as “text in image”.
- [ ] Odd bullet colours that follow surrounding text are fixed together with text to keep visual coherence.

8. Alt text and handwriting

Every image appears on the alt-text review page with an AI draft, nearby caption or paragraph, and flags. Nothing is auto-approved; human approval is required before writing into files.

ALT-TEXT REVIEW

COLUMN	WHAT IT SHOWS
Preview	Cropped image, page thumbnail, size, type (photo/diagram/chart)
AI draft	Qwen2.5-VL 7B draft, nearby text context (caption, preceding paragraph), confidence
Flags	Text in image, colour-only meaning, long description candidate (> 150 chars needed), decorative candidate
Actions	Approve / edit / mark decorative / Approve all remaining / Draft with AI (re-run)

Saving writes alt text into DOCX / PPTX / PDF (StructTreeRoot / Alt entry). PDFs are regenerated with updated structure tree; Word and PowerPoint updated in place. Ollama runs locally with GPU acceleration; no image leaves the host machine.

HANDWRITING

Drawing detector distinguishes hand-drawn figures from handwritten text. If handwriting is detected:

- Model transcribes page with attention to math symbols and chemical notation.
- Typed version built with real Word equations (OMML) via pandoc + matplotlib LaTeX → OMML conversion, preserving layout proximity to original.
- Tagged PDF generated from typed version with headings, paragraphs, and alt text for drawings.
- Operator sees side-by-side: original scan vs. typed version, can edit transcription, approve typed version, or keep original and mark as “handwritten — transcribed”.

Limit: model guesses when handwriting is ambiguous; report lists low-confidence tokens.

9. Logic checks

Beyond syntax, the Workbench runs semantic logic checks that catch patterns common in course materials.

CHECK	WHAT IT DOES	REPORT OUTPUT
Headings start at H1, never skip	Scans outline, repairs H1 missing, H1→H3 jumps collapsed to H1→H2	Fixed count + remaining skips flagged with page + heading text
Layout tables become text	Detects tables used for positioning (no header row, uneven columns) → converts to paragraphs	List of converted tables, former location
Data tables without headers	First row marked header if not marked; merged header cells preserved	Fixed count + tables still without clear headers flagged
Figures without alt	Every figure requires Alt or Artifact; custom roles resolved	Count + link to alt-text review
Unlabeled form fields	PDF AcroForm fields checked for accessible name (TU or tooltip); Word form controls checked	Flagged list — not auto-labeled
Reading order jumps	Flags when PDF drawing order jumps across columns or from footer to body	Flagged, not reordered automatically — human verifies
Video/audio flagged	Detects embedded media or linked media text → flags for captions/transcript	Flagged list with timestamps/links
Invisible text	Text with rendering mode 3, white on white, or outside crop box flagged	List with reason and location
Odd bullet colours follow text	When bullet colour differs but text colour is primary, bullet recoloured to match text	Fixed silently, noted in report

Reports: `remediation-report.html` (human-readable) and `remediation-report.csv` (machine-readable for tracking). Both include fixed, flagged, and skipped counts per file.

10. What it cannot do — limits

Some accessibility requirements require human judgement. The Workbench fixes what can be safely automated and flags the rest.

AREA	LIMIT	WHAT A PERSON DOES
Alt text	AI drafts can be wrong, miss context, or describe decorative images as informative	Review every AI draft on alt-text page; edit for accuracy, mark decorative where appropriate; add long descriptions for complex diagrams in surrounding text
Handwriting	Model guesses when handwriting is faint, cursive, or mixed with drawings; math symbols can be misread	Verify transcription side-by-side; correct equations; approve only when meaning preserved
OCR	Math, small print, poor scans, coloured paper, and hand-annotated pages have lower accuracy	Check low-confidence words listed in report; re-scan at 300dpi if needed; add corrections to text-version
Math in typed PDFs	Equations in born-digital PDFs are not converted to MathML; tagged as Figure with alt or as P with readable alt	For STEM courses, provide source Word/LaTeX or add MathML description in alt text; consider rebuilding through Word for better structure
Drawn graphics	Charts that are vector shapes (not images) have no alt by default; data not extractable	Add alt text describing trend/data; or replace chart image with data table + alt
Reading order	Order is set from the order PDF draws text; flagged when suspicious but not reordered automatically to avoid breaking layout	Spot-check reading order in Acrobat or screen reader; reorder manually if needed
Tables in PDFs	Tag-in-place does not create table tags (TH/TD) because that would require redrawing; structure preserved as P/List where possible	For data-heavy PDFs, use Rebuild through Word option to generate true table tags, or provide HTML/Word equivalent
Forms	Unlabeled AcroForm fields flagged, not auto-labeled (label requires understanding intent)	Add accessible names in Acrobat, or recreate form in accessible tool
Media	Captions, transcripts, and audio descriptions not generated	Upload captioned version or add transcript file; Workbench flags presence only
Invisible leftover text	Some PDFs contain hidden duplicate text (e.g., from failed OCR); flagged but not auto-removed if it may affect visual match	Review invisible text list; remove in Acrobat if safe
PowerPoint tables	Table header flagged, not set automatically via UI because	Set header row in PowerPoint: Table Design → Header Row, or add alt text describing headers

AREA	LIMIT	WHAT A PERSON DOES
	PowerPoint object model lacks header flag	
Scale	One laptop, 2 files at a time; large scanned PDFs (200+ pages) may take several minutes per file	Run batches overnight; use RETENTION_DAYS to auto-clean; split very large PDFs if possible
Plain language / keyboard / colour-only	Cannot be judged automatically — requires human understanding of meaning and interaction	Use plain language checklist, keyboard test, and colour-only review alongside Workbench

11. Installation

Option A — Docker (recommended) — Windows, Mac, Linux

1. Install Docker Desktop. On Windows, enable WSL2 integration in Docker Desktop → Settings → Resources → WSL Integration.
2. Clone or unzip the Workbench folder. Edit `docker-compose.yml` and set `APP_PASSWORD` to a strong value. This password protects the web UI.
3. Run `docker compose up -d --build` from the project root. Wait for build (first time 5–10 minutes).
4. Open `http://localhost:8080` in browser. Log in with the password you set. Teammates on same network can use your machine name or IP: `http://MACHINE-NAME:8080` or `http://192.168.x.x:8080`.
5. Optional: install Ollama on host for alt-text and handwriting. See below.

Data stays in Docker volume or bind-mounted `./data`. No files leave host.

Option B — Run directly — Linux / macOS

Use when Docker is not allowed or for development.

UBUNTU / DEBIAN

```
sudo apt update
sudo apt install tesseract-ocr ghostscript
qpdf poppler-utils libreoffice-writer
pandoc
```

MACOS

```
brew install tesseract ghostscript qpdf
poppler pandoc
brew install --cask libreoffice
```

RUN

```
python3 -m venv .venv && source
.venv/bin/activate
pip install -r requirements.txt
HOST=0.0.0.0 APP_PASSWORD=change-me python
app.py
```

Windows without Docker: Use WSL2 Ubuntu, then follow Linux steps inside WSL. Access via `http://localhost:8080` from Windows browser.

SETTINGS — ENVIRONMENT VARIABLES

SETTING	DEFAULT	NOTES
`HOST / PORT`	0.0.0.0 / 8080	Bind address. Use 127.0.0.1 for local-only, 0.0.0.0 for LAN access.
`APP_PASSWORD`	(required)	Protects UI. Set in docker-compose.yml or env. No default — must be set.
`WORKERS`	2	Parallel jobs. 1 for low RAM, 2 for typical laptop.
`RETENTION_DAYS`	14	Auto-delete job folders. History entry remains.
`MAX_UPLOAD_MB`	1000	Per file limit. Larger files rejected with message.
`DOC_LANG`	en-US	Default document language if none detected.
`OCR_LANG`	eng	Tesseract languages, e.g., eng+spa.
`CHECK_LINKS`	false	When true, fetches URLs to verify. Only optional network call.
`CONVERT_LISTS`	true	Convert hand-typed bullets to real lists.

SETTING	DEFAULT	NOTES
`VERAPDF`	(optional path)	If veraPDF installed, validates PDF/UA after remediation.
`SOFFICE`	libreoffice	Path to LibreOffice soffice binary for conversions.
`OLLAMA_URL / MODEL`	http://host.docker.internal:11434 / qwen2.5vl:7b	Alt-text and handwriting. Install Ollama on host, run `ollama pull qwen2.5vl:7b`. GPU preferred.

Optional veraPDF: Add veraPDF container or binary and set VERAPDF path to validate PDF/UA compliance after each job. Report includes veraPDF summary if enabled.

Optional alt-text with Ollama: Install Ollama on host (not in Docker). On Windows/Mac, Docker can reach host via host.docker.internal. Pull model once; first alt-text request loads model into GPU VRAM.